



Human Technopole

Nextflow zero2hero

28 January 2026

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Nextflow: Reporting and Monitoring



Summary

- Resource and time tracking
- Configure and use Tower to monitor execution

Nextflow: Reporting and Monitor

Resource and time tracking

Nextflow includes built-in tools for resource profiling.

A report can be requested by adding the option **-with-report <filename>.html** to the command line.

This HTML report allows users to analyze the utilization of resources (CPU and memory) by each process.

To optimize performance, it is vital to analyze the utilization results **as a percentage of what was allocated**,.

The profiling report helps users right-size resource allocations based on actual data rather than guesswork.

Summary

The **Summary** section reports the execution status, the launch command, overall execution time, and other workflow metadata.

 Nextflow Report Summary Resources Tasks [deadly_baekeland]

Nextflow workflow report

[deadly_baekeland]

Workflow execution completed successfully!

Run times
09-Dec-2025 02:43:21 - 09-Dec-2025 02:46:58 (duration: **3m 37s**)

4 succeeded

Nextflow command

```
nextflow run rnaseq-nf --profile conda --with-report report.html
```

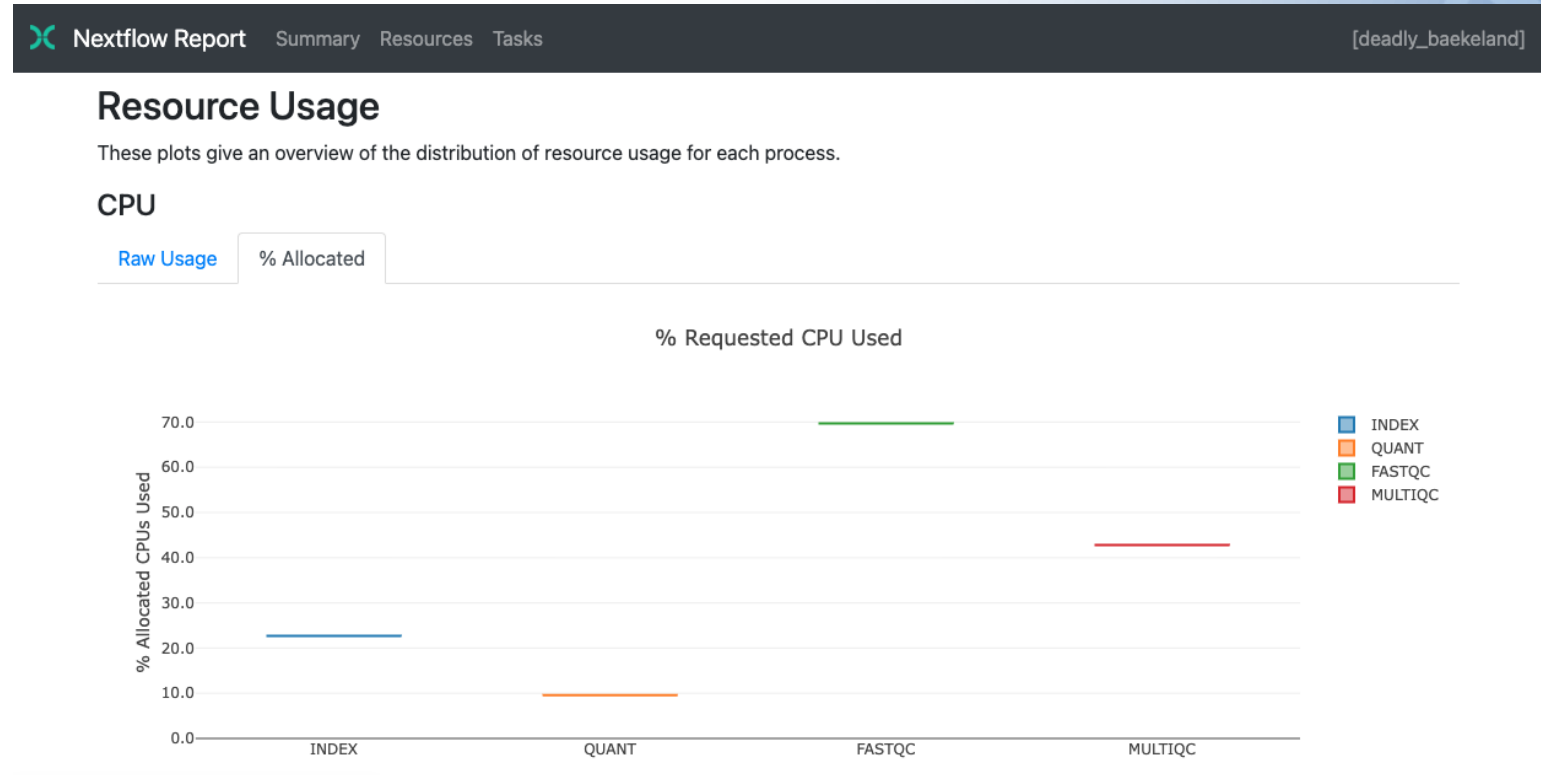
CPU-Hours	(a few seconds)
Launch directory	/workspaces/sandbox
Work directory	/workspaces/sandbox/work
Project directory	/home/gitpod/.nextflow/assets/nextflow-io/rnaseq-nf
Script name	main.nf
Script ID	295b10c8d2badd387ed3f6fc5e56e45e
Workflow session	61a4b8bc-9051-4108-b0b2-00f8ccd83e55
Workflow repository	https://github.com/nextflow-io/rnaseq-nf , revision master (commit hash ca20a6dfd2d799b903e557dc2e736d55419c7a1a)
Workflow profile	conda
Nextflow version	version 25.04.6, build 5954 (01-07-2025 11:27 UTC)

Resource Usage

The Resources section plots the distribution of resource usage for each workflow process using the interactive *plotly.js* plotting library.

Plots are shown for CPU, memory, job duration, and disk I/O. Each usage metric is shown both in raw form and as a percentage of the requested resources.

These plots are helpful to check that task resources are used efficiently.



Task

The **Tasks** section lists all executed tasks, reporting the status, command script, and other metrics for each task.

 Nextflow Report [Summary](#) [Resources](#) [Tasks](#) [deadly_baekeland]

Tasks

This table shows information about each task in the workflow. Use the search box on the right to filter rows for specific values. Clicking headers will sort the table by that value and scrolling side to side will reveal more columns.

Values shown as: Human readable

Show 25 entries

Filter: Metrics Metadata All Search:

task_id	process	tag	status	hash	allocated cpus	%cpu	allocated memory	%mem	vmem
1	RNASEQ:INDEX	ggal_1_48850000_49020000	COMPLETED	54/12b239	1	22.7	-	0.2	52.016 MB
2	RNASEQ:FASTQC	ggal_gut	COMPLETED	c7/e107f8	1	69.7	-	2.4	3.002
3	RNASEQ:QUANT	ggal_gut	COMPLETED	83/9c395b	1	9.6	-	0.6	368.9 MB
4	MULTIQC	-	COMPLETED	94/c235e1	1	42.8	-	1.4	571.58 MB

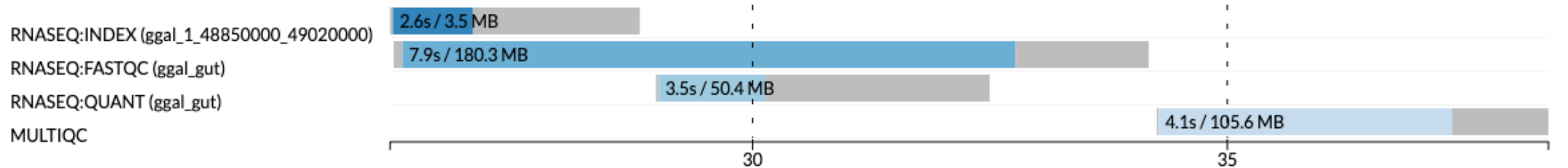
Execution timeline

Processes execution timeline

Launch time: 09 Dec 2025 16:07

Elapsed time: 15.8s

Legend: job wall time / memory usage (RAM)



The x-axis shows elapsed time relative to workflow launch. The y-axis shows a bar for each task execution. Each bar has the following properties (with the equivalent trace fields in parentheses):

- Bar length: Task duration (**duration**)
- Bar color: Identifies tasks from the same process
- Colored area: Real execution time (**realtime**)
- Gray area (left): Task scheduling time (**start - submit**)
- Gray area (right): Task termination time (**complete - start - realtime**)
- Label: Task duration (**duration**) and peak memory usage (**peak_rss**)

Nextflow: Reporting and Monitor

Configure and use tower to monitor execution

Seqera Platform, previously known as Nextflow Tower, is the centralized command post for data management and workflows. It brings monitoring, logging and observability to distributed workflows and simplifies the deployment of workflows on any cloud, cluster or laptop.

In Seqera Platform terminology, a workflow is what we've been working on so far, and pipelines are pre-configured workflows that can be used by all users in a workspace.

It is composed of a workflow repository, launch parameters, and a compute environment.

We'll stick to these definitions in this section.

Seqera core features include:

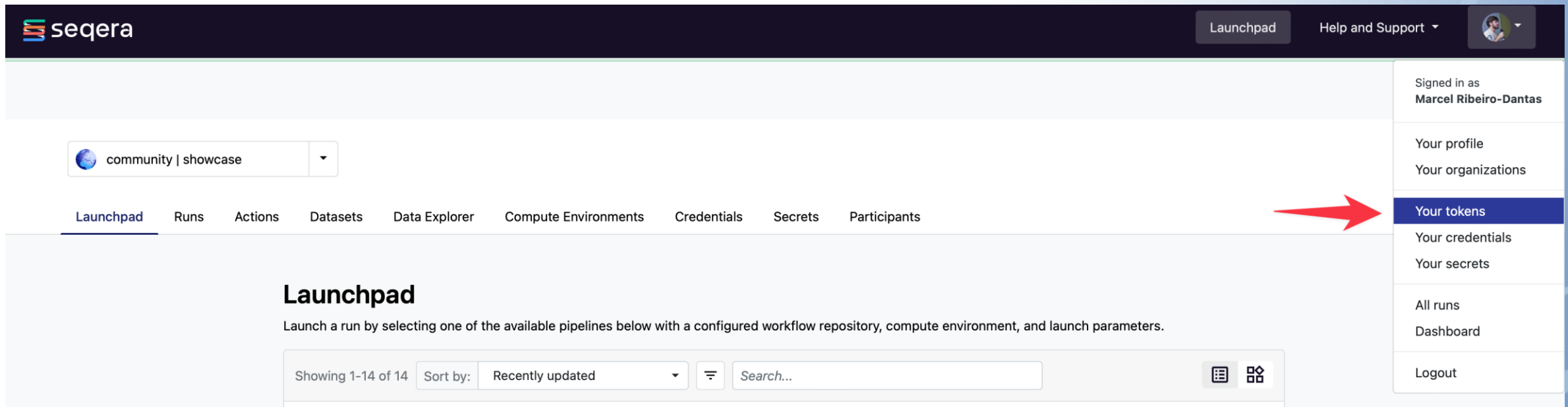
- The launching of pre-configured pipelines with ease.
- Programmatic integration to meet the needs of an organization.
- Publishing pipelines to shared workspaces.
- Management of the infrastructure required to run data analysis at scale.

Nextflow: Reporting and Monitor

Configure and use tower to monitor execution

1. Create a new token

You can access your tokens from the **Settings** drop-down menu:

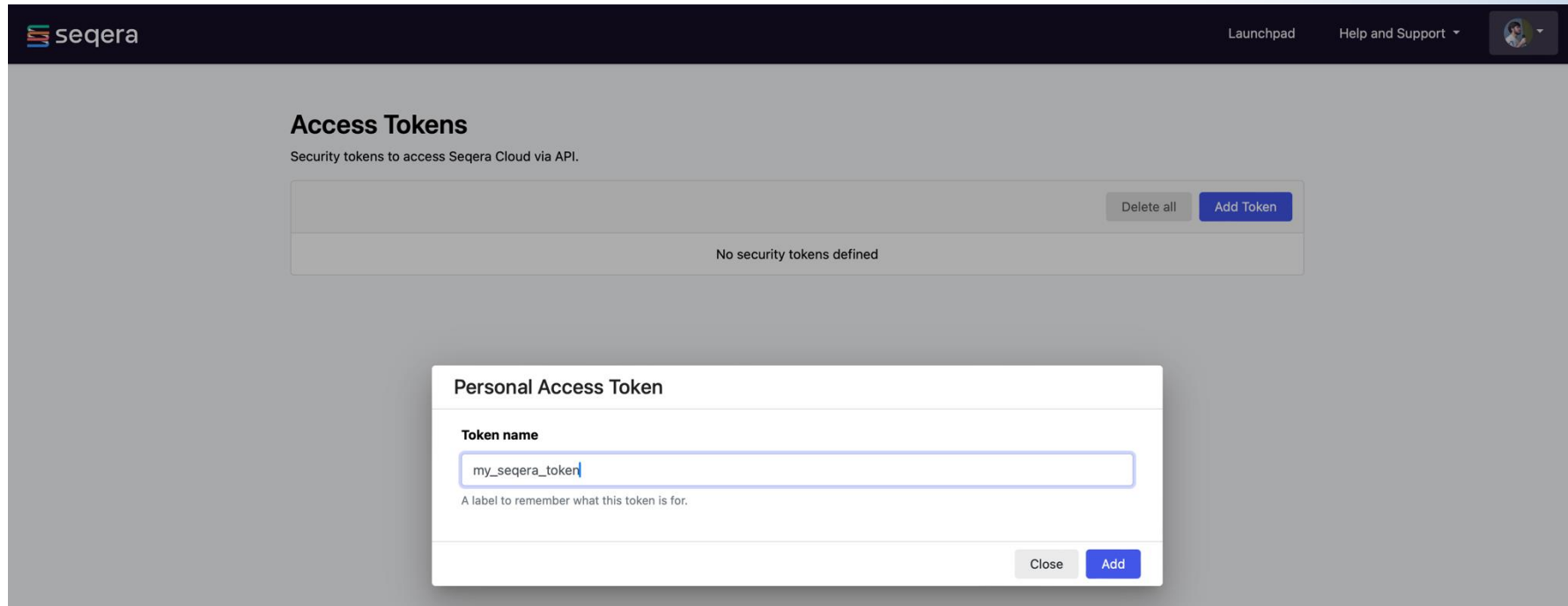


The screenshot shows the Seqera Launchpad interface. At the top, there's a dark blue header with the Seqera logo on the left, 'Launchpad' and 'Help and Support' in the center, and a user profile icon on the right. Below the header, there's a navigation bar with tabs: 'Launchpad', 'Runs', 'Actions', 'Datasets', 'Data Explorer', 'Compute Environments', 'Credentials', 'Secrets', and 'Participants'. A red arrow points to the 'Your tokens' option in the user profile dropdown menu. The main content area is titled 'Launchpad' and contains a description: 'Launch a run by selecting one of the available pipelines below with a configured workflow repository, compute environment, and launch parameters.' Below this, there's a filter section with 'Showing 1-14 of 14', 'Sort by: Recently updated', and a search bar. The user profile dropdown menu is open, showing options: 'Signed in as Marcel Ribeiro-Dantas', 'Your profile', 'Your organizations', 'Your tokens' (highlighted with a red arrow), 'Your credentials', 'Your secrets', 'All runs', 'Dashboard', and 'Logout'.

Nextflow: Reporting and Monitor

Configure and use tower to monitor execution

2. Name your token



The screenshot shows the Seqera web interface. At the top, there's a dark header with the Seqera logo, "Launchpad", "Help and Support", and a user profile icon. The main content area is titled "Access Tokens" with the subtitle "Security tokens to access Seqera Cloud via API.". Below this, there's a button "Delete all" and a blue button "Add Token". A message states "No security tokens defined". A modal window titled "Personal Access Token" is open, featuring a "Token name" label and a text input field containing "my_seqera_token". Below the input field, it says "A label to remember what this token is for.". At the bottom of the modal are "Close" and "Add" buttons.

seqera

Launchpad Help and Support

Access Tokens

Security tokens to access Seqera Cloud via API.

Delete all Add Token

No security tokens defined

Personal Access Token

Token name

my_seqera_token

A label to remember what this token is for.

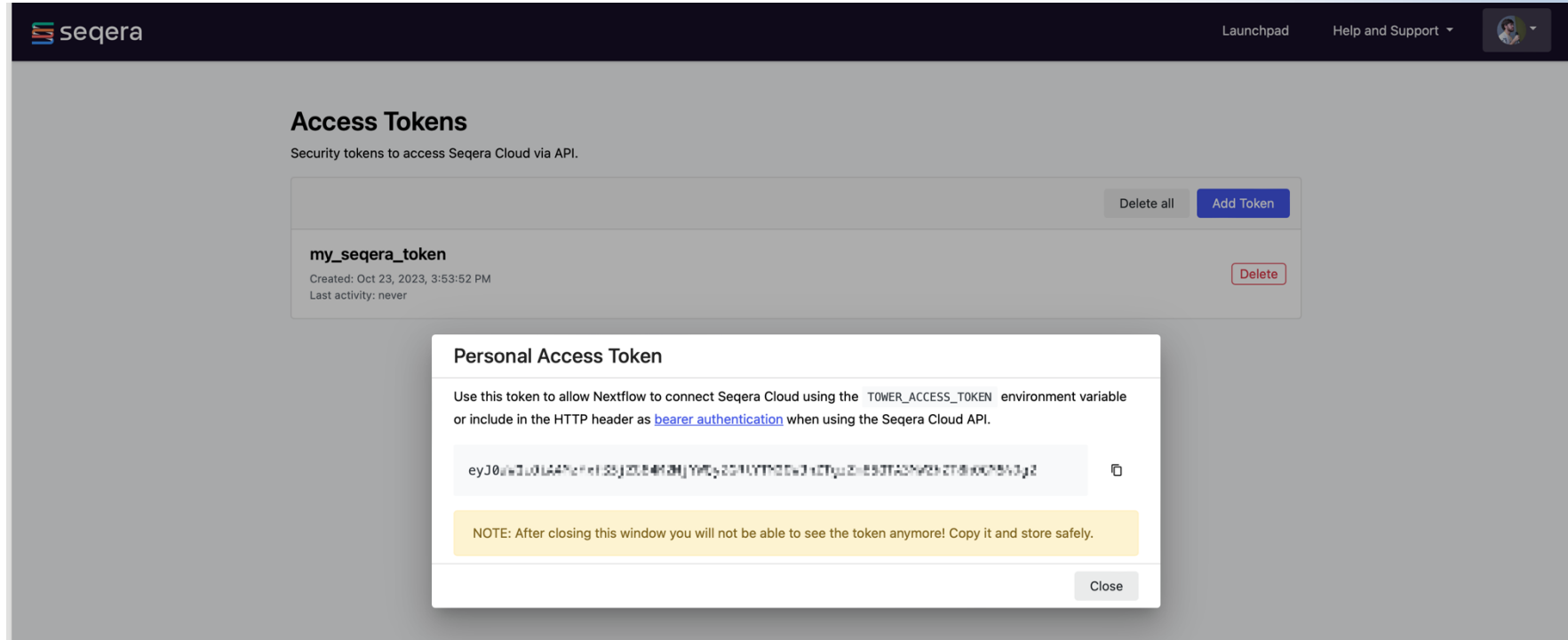
Close Add

Nextflow: Reporting and Monitor

Configure and use tower to monitor execution

3. Save your token safely

Copy and keep your new token in a safe place.



The screenshot shows the Seqera web interface. At the top, there's a dark header with the Seqera logo, "Launchpad", "Help and Support", and a user profile icon. The main content area is titled "Access Tokens" with a subtitle "Security tokens to access Seqera Cloud via API.". Below this, there's a table of tokens. One token is listed: "my_seqera_token", created on "Oct 23, 2023, 3:53:52 PM", with "Last activity: never". To the right of the token name are buttons for "Delete all" and "Add Token". To the right of the token details is a "Delete" button. A modal window titled "Personal Access Token" is open in the foreground. It contains instructions: "Use this token to allow Nextflow to connect Seqera Cloud using the `TOWER_ACCESS_TOKEN` environment variable or include in the HTTP header as [bearer authentication](#) when using the Seqera Cloud API." Below the text is a text box containing a long alphanumeric token string, followed by a copy icon. A yellow note box says: "NOTE: After closing this window you will not be able to see the token anymore! Copy it and store safely." At the bottom right of the modal is a "Close" button.

Nextflow configuration Practices



Now let's see how this works in practice

Thank you.

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